

Ideation Phase
Brainstorm & Idea Prioritization Template

Date	1 March 2026
Team ID	699d5f82fdf409fdbd1bec66
Project Name	Malaria Detection Using Deep Learning
Maximum Marks	4 Marks

Brainstorm & Idea Prioritization:

Step-1: Team Gathering, Collaboration and Select the Problem Statement

The project team conducted a brainstorming session to discuss healthcare problems that could be solved using Artificial Intelligence and Machine Learning. During the discussion, several problem areas were identified including disease detection, medical image analysis, and automated diagnostics.

After evaluating different healthcare challenges, the team selected the problem of slow and manual malaria diagnosis using microscopic blood smear analysis. Manual diagnosis requires trained pathologists and takes significant time, which can delay treatment, especially in rural or resource-limited areas.

Therefore, the team decided to focus on developing an AI-based malaria detection system using deep learning that can automatically classify blood smear images as Parasitized or Uninfected.

The selected problem statement is:

“Develop an automated system using deep learning to detect malaria parasites from microscopic blood smear images to assist doctors and laboratory technicians in faster and more accurate diagnosis.”

Step-2: Brainstorm, Idea Listing and Grouping

During the brainstorming session, the team generated multiple ideas related to improving malaria diagnosis and healthcare automation.

Idea ID	Proposed Idea
I1	Manual malaria detection assistance software
I2	Image processing tool for blood smear analysis
I3	AI-based malaria parasite detection using CNN
I4	Mobile application for malaria detection
I5	Web-based system for uploading blood smear images and detecting malaria
I6	Automated medical image classification system
I7	Cloud-based malaria detection platform

Idea Grouping

The generated ideas were grouped into categories:

Category 1: Manual Assistance Systems

- Manual malaria detection software
- Image processing assistance tool

Category 2: AI-Based Detection Systems

- Deep learning malaria detection system
- Automated medical image classification

Category 3: Platform-based Solutions

- Web-based malaria detection system
- Mobile-based malaria detection application
- Cloud-based diagnostic system

After grouping the ideas, the team evaluated which concept was most practical, feasible, and impactful.

Step-3: Idea Prioritization

The ideas were evaluated based on the following criteria:

- Technical feasibility
- Real-world impact
- Implementation complexity
- Availability of dataset
- Scalability

Idea	Feasibility	Impact	Complexity	Priority
Manual malaria detection software	Medium	Medium	Low	Low
Image processing tool	Medium	Medium	Medium	Medium
AI-based malaria detection using CNN	High	High	Medium	High
Mobile malaria detection app	Medium	High	High	Medium
Web-based malaria detection system	High	High	Medium	High
Cloud-based detection platform	Medium	High	High	Medium